From Data Cleaning to Anomaly Intelligence

Transforming raw substation data into an anomaly-ready dataset

Sense Reply x Albert School – BDD Challenge

Phase 1 Recap: Stabilizing the Dataset

Missing values fixed

Duplicate rows removed

10-minute frequency enforced

Dataset is now structurally stable

Why Cleaning Wasn't Enough

- Substations follow daily cycles
- Human activity drives weekly seasonality
- Anomalies must be judged in context
- → Clean data ≠ intelligent data

We needed to teach the dataset how to understand itself.

Overview of Engineered Features ("Intelligence Layer")

We added features that provide:

Temporal context

(hour, weekday, weekend)

Behavioral variability

(rolling std)

Risk signals

(strict & relaxed anomaly flags)

Feature: Hour, Day, Weekday, Weekend

Why we added them:

- Capture predictable daily peaks
- Distinguish weekday vs weekend behaviour
- Provide the model with time-aware expectations

Code:

```
df['hour'] = df.index.hour
```

Feature: Rolling Standard Deviation (Instability Detector)

Why we added them:

- Detect local volatility
- Identify early signs of instability
- Highlight unusual sensor behaviour

Code:

```
df['std_rolling'] = df['consumption'].rolling(3).std()
```

Strict vs Relaxed Anomaly Flags

Strict flag (Z > 3)

- Very rare
- High-priority anomalies

Relaxed flag (Z > 2)

- More frequent
- Early-warning signals

Code:

```
df['anomaly_relaxed'] = z_scores > 2
```

Connecting to Real Architecture (Business Case)

Origins of problems:

SCADA polling delay

→ missing values

Gateway retransmission

→ duplicates

Clock drift / congestion

→ irregular intervals

Sensor instability

→ volatile consumption

Our engineered features act as operational diagnostics.

Before vs After (Visual Transformation)

Left = raw dataset:

- inconsistent spacing
- missing timestamps
- noisy readings
- no contextual information

Right = final dataset:

- perfect 10-min spacing
- contextual features added
- anomalies highlighted

A Day in the Life of a Substation

Morning Peak

6 AM - 9 AM

Consumption: Rises Sharply

As activities begin, energy demand rapidly increases.

Afternoon Plateau

9 AM - 5 PM

Consumption: Steady High

Workplaces and commercial operations maintain high usage.

Evening Rise

5 PM - 8 PM

Consumption: Secondary Peak

People return home, driving a second surge in demand.

Night Low

8 PM - 6 AM

Consumption: Minimal

Essential systems operate, with overall demand at its lowest.

The engineered features (hour, weekday, rolling std) teach the model to expect this rhythm. When consumption breaks the pattern → anomaly detected.

A Smarter Substation: From Data to Decisions

From reactive monitoring to predictive intelligence

Stabilized the Dataset

Fixed missing values, removed duplicates, enforced 10-min frequency

Added Intelligence Layer

Engineered temporal, behavioral, and risk features

Built Dual Detectors

Strict (Z>3) and relaxed (Z>2) anomaly flags

Connected to Real Pipeline

Linked SCADA → IoT → Cloud architecture

We didn't just clean data; We prepared the grid for smarter decisions.